Computer Graphics and Man-Machine Interface

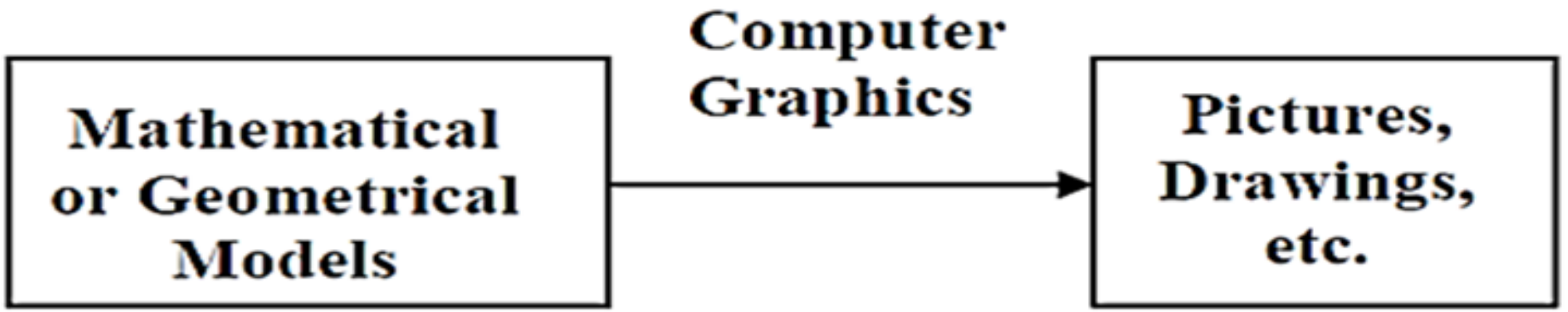
Dr. Manar Donia

CMPn322 - CMPN322 - CMP 422

What is Computer graphics?

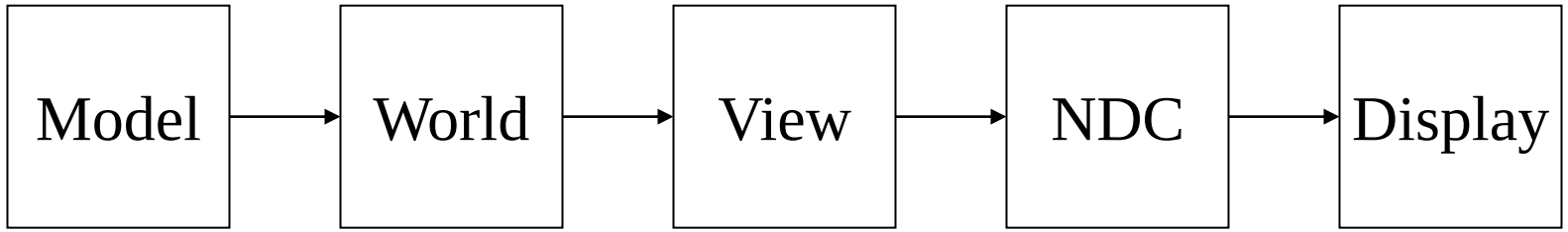
- Computer graphics means creation, storage and manipulation of models and images.



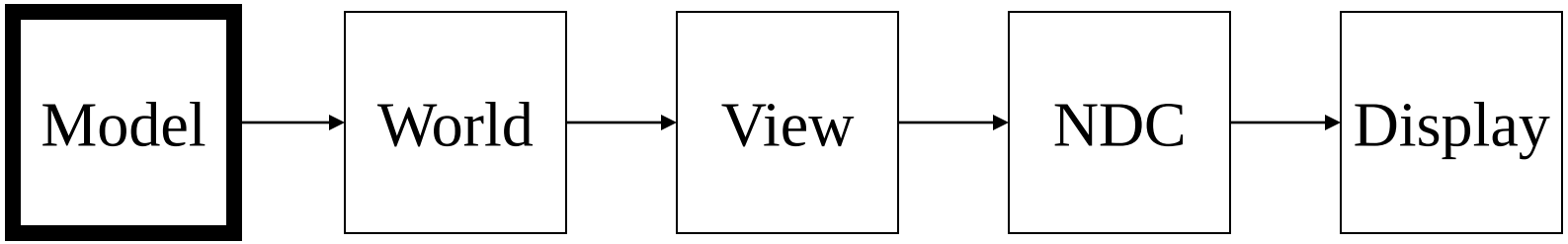


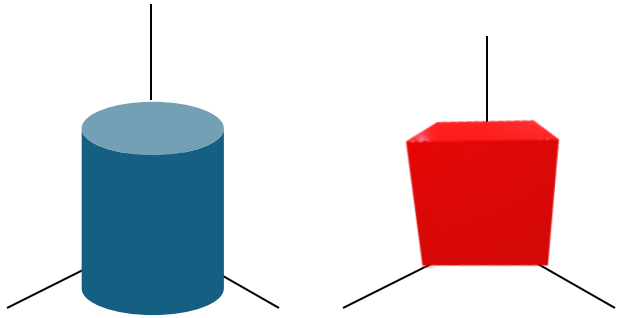
From model to image

Start from the model until it is displayed
Coordinates and transformations



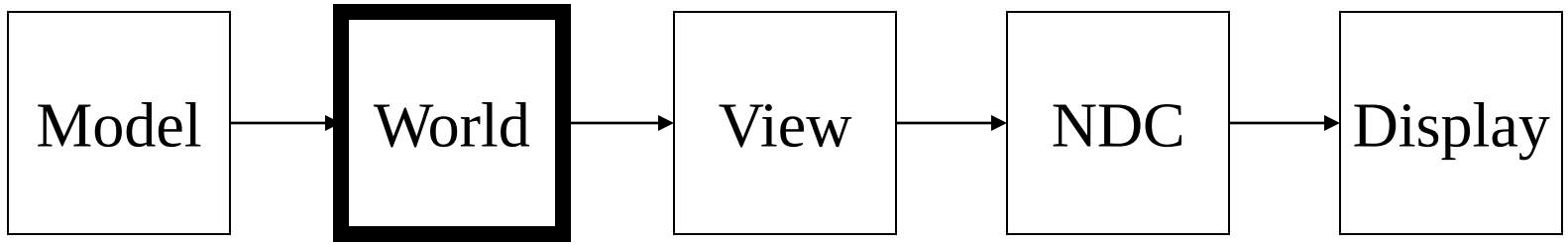
From model to image

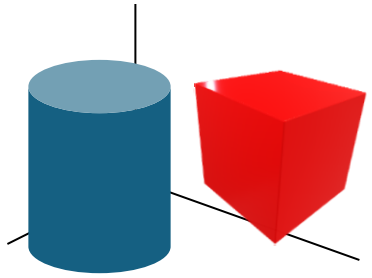




Local or modeling coordinates

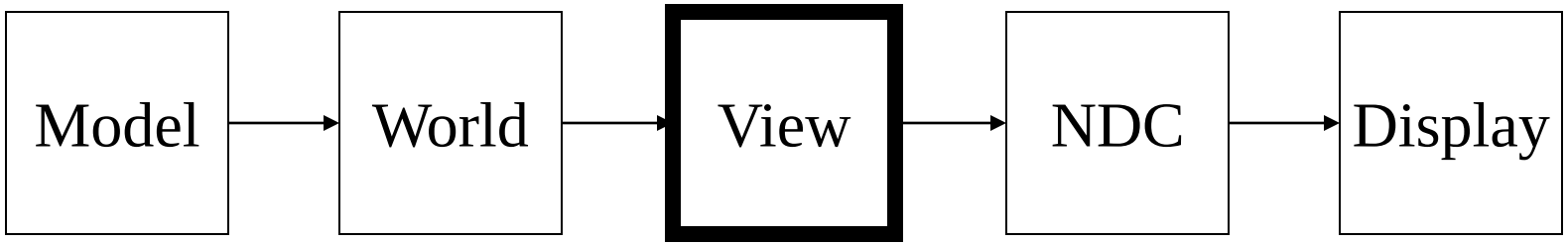
From model to image

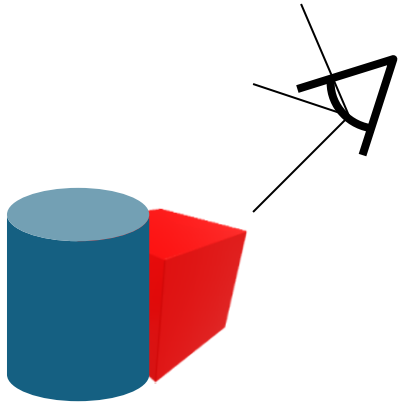




Position models in scene: World coordinates

From model to image



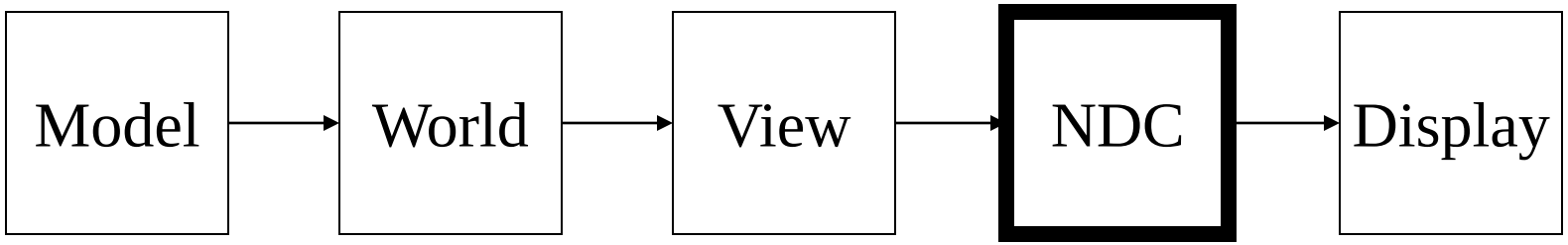


project on virtual screen

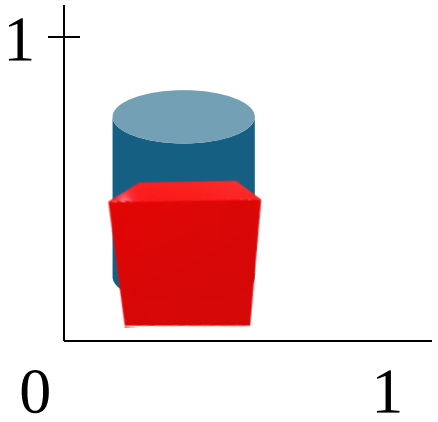
Viewing coordinates

Visible surfaces, shading

From model to image

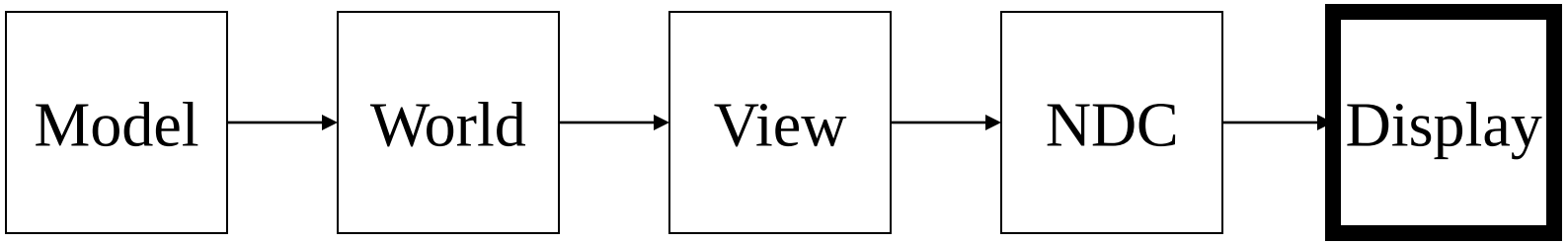


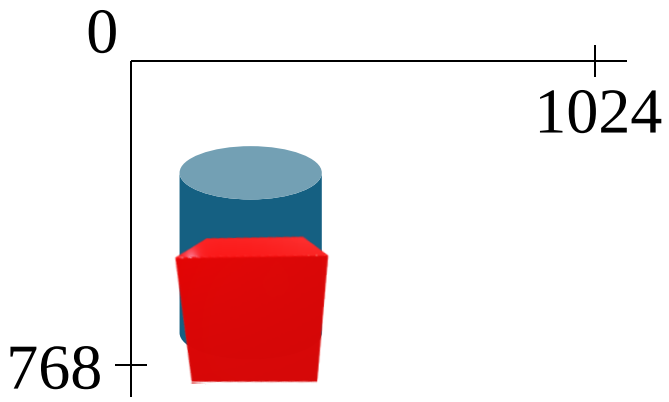
Display:



Normalized Device Coordinates

From model to image





Display on screen:

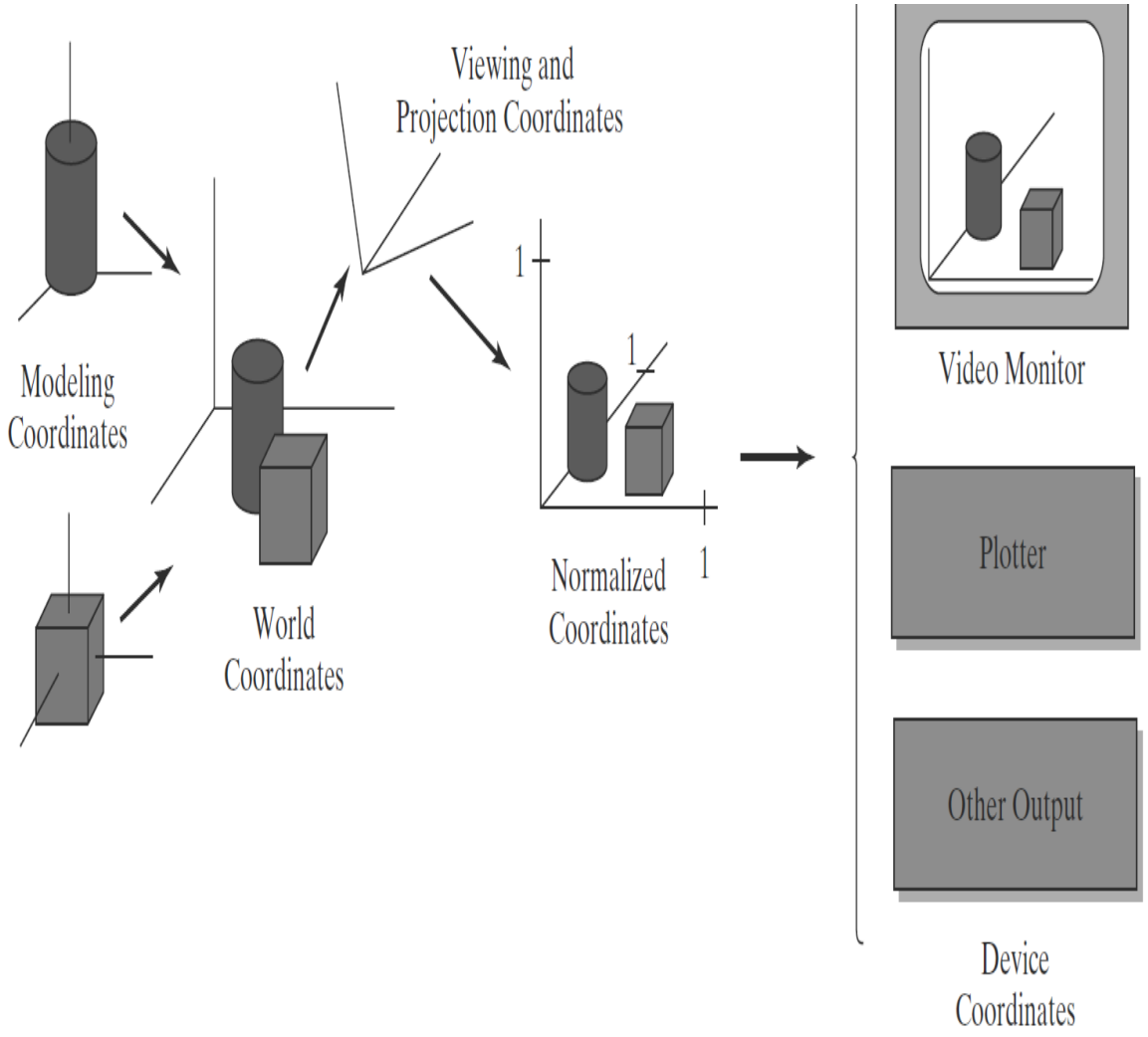
Device Coordinates

Interaction

From model to image

The process of transforming coordinates from modeling to device coordinates for a three-dimensional scene involves several steps.

1. object shapes are defined within their own modeling-coordinate systems.
2. These shapes are then placed within a global world-coordinate scene.
3. The world-coordinate data is processed through the viewing pipeline, converting it into viewing and projection coordinates, and further into normalized coordinates.
4. Finally, device drivers take the normalized-coordinate representation of the scene and send it to the output devices for display.



Elements of Pictures Created in Computer Graphics.

Output primitives→ The basic geometric structure used to describe scenes

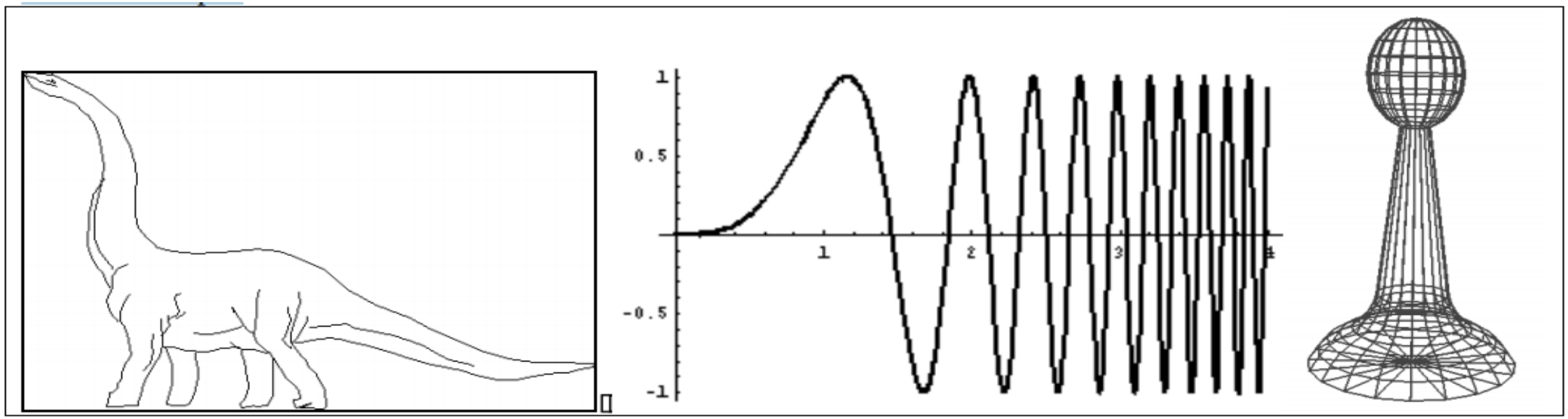
- Polyline
- Text
- filled regions
- raster images

Each elements has set of attributes

- The **attributes** of a graphic primitive are the characteristics that affect how it appears, such as color , style, and thickness.

Elements of Pictures Created in Computer Graphics

Polyline **(**the pictures made up of polylines are called **line drawing)**

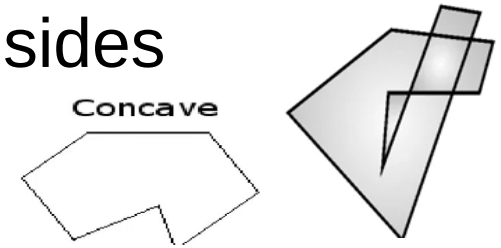


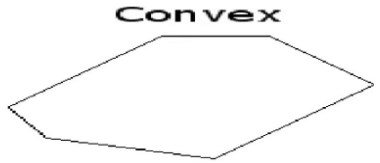
- a polyline drawing of a dinosaur**.**
- a plot of a mathematical function.
- a wireframe rendering of a 3D object.

Elements of Pictures Created in Computer Graphics

- Polylines
  - Connected sequence of straight lines
  - Pictures made up of polylines
    - Each line is called an edge
    - Two adjacent lines meet at vertex
  - Polygon is closed polyline where its start point equal to its last point
    - Simple polygon → has a well defined two regions(inside and outside)
    - Complex polygon → has a self-intersecting sides
    - Concave polygon→ if it is curved inward
    - Convex polygon→ if it is curved outward

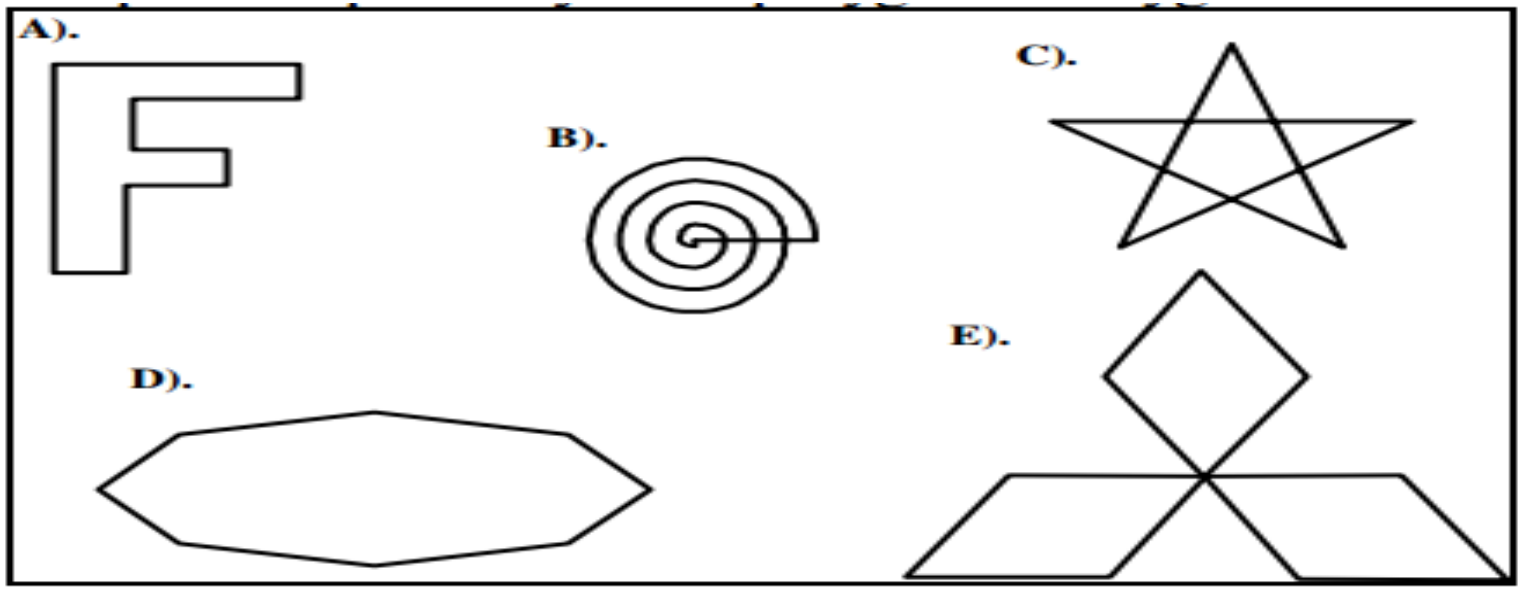






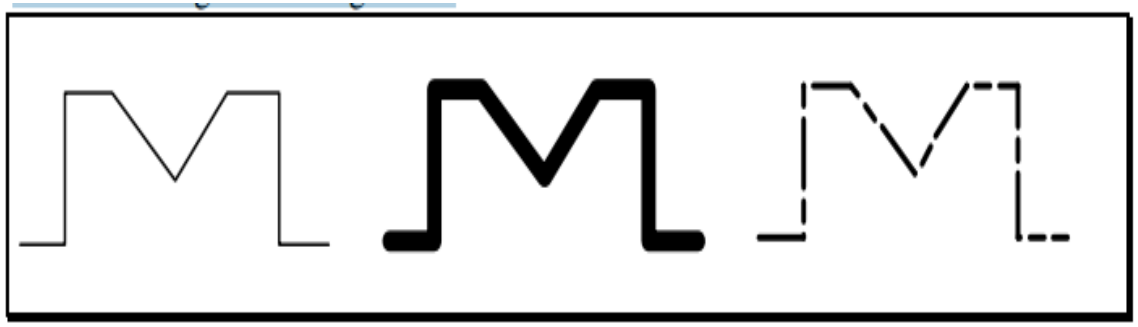
Elements of Pictures Created in Computer Graphics

- Polylines are specified as a list of vertices, each given by a coordinate pair: (*x*0, *y*0), (*x*1, *y*1), (*x*2, *y*2), ..., (*x*n, *y*n)
- A special case arises when a line segment shrinks to a single point, and is drawn as a “dot” drawDot(x1, y1);



Elements of Pictures Created in Computer Graphics

- Attributes of Lines and Polylines
  - Color
  - Thickness
  - Type(dashed, solid, dotted)
  - Way edges are blended



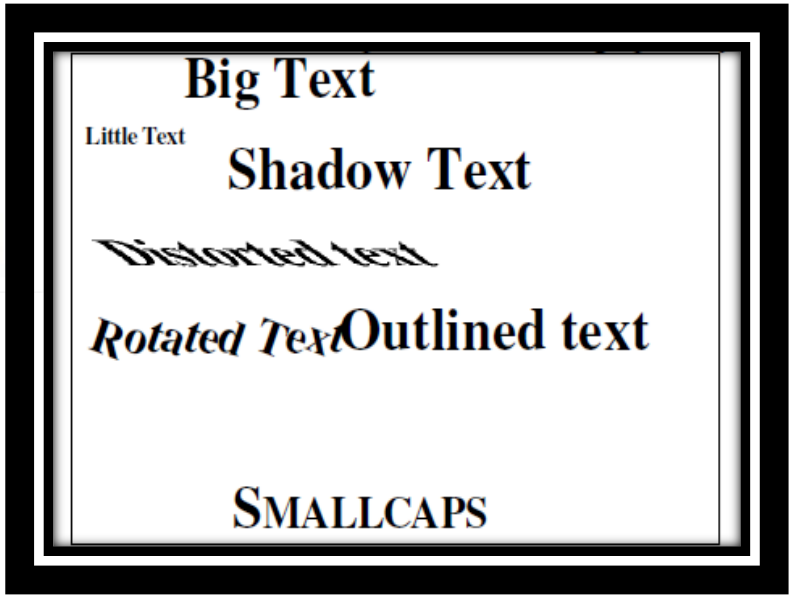


Non joint | Mitered | Rounded | Beveled

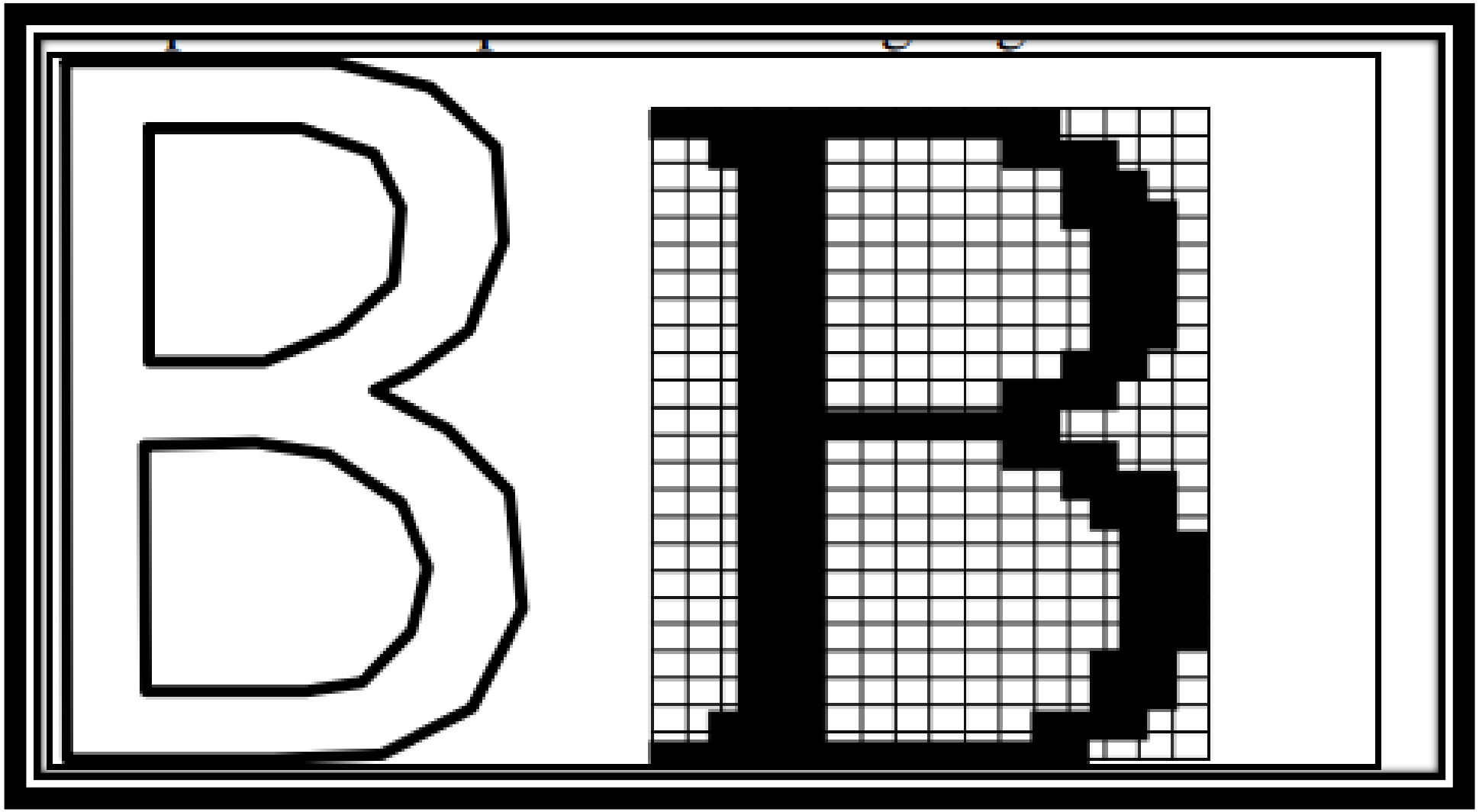
Elements of Pictures Created in Computer Graphics

- Text
  - A tool to draw a character string might look like: drawString(x, y, string); It places the starting point of the string at position (*x*, *y*), and draws the sequence of characters stored in the variable string.
  - Attributes:
    - Font face
    - Weight
    - Size
    - Style
    - Spacing
    - Orientation

Face (family)	Weight	Style (slant)	Size
Helvetica	Normal	Normal	8 point
Times–Roman	**Bold**	*Italic*	10 point
Courier			12 point

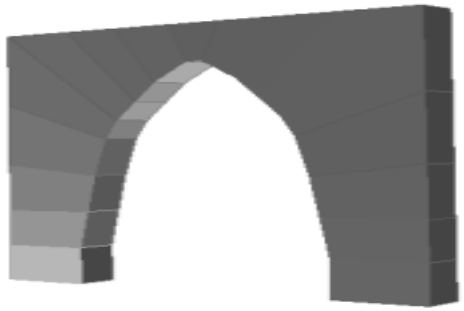


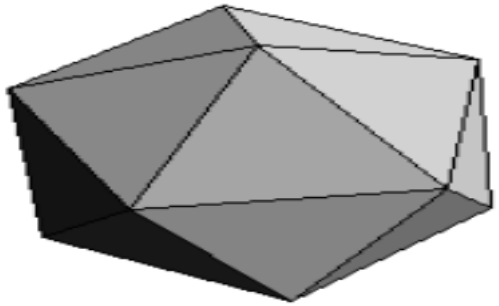
A character shape defined by a polyline or by a arrangement of dots



Elements of Pictures Created in Computer Graphics

- **Filled Region**
  - Shape filled with color
  - Boundary of filled shape is a polygon
  - To draw a filled polygon, one would use a routine like: fillPolygon(poly, pattern);
- Attributes:
  - Enclosing border
  - Pattern
  - color

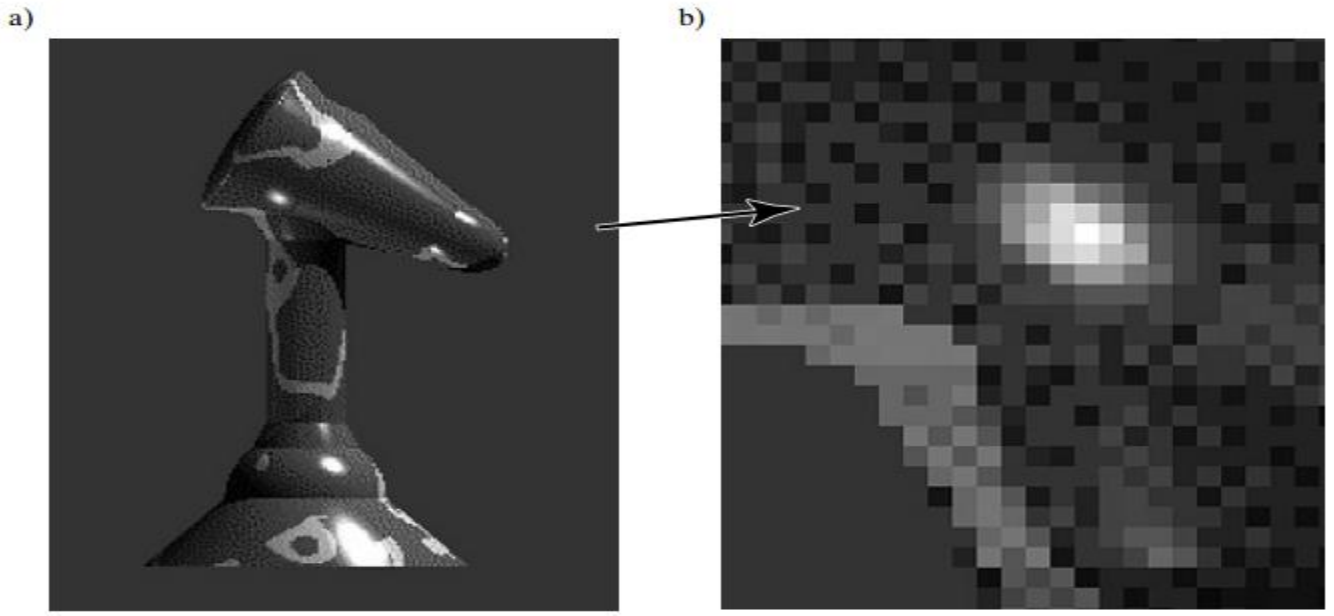




Elements of Pictures Created in Computer Graphics

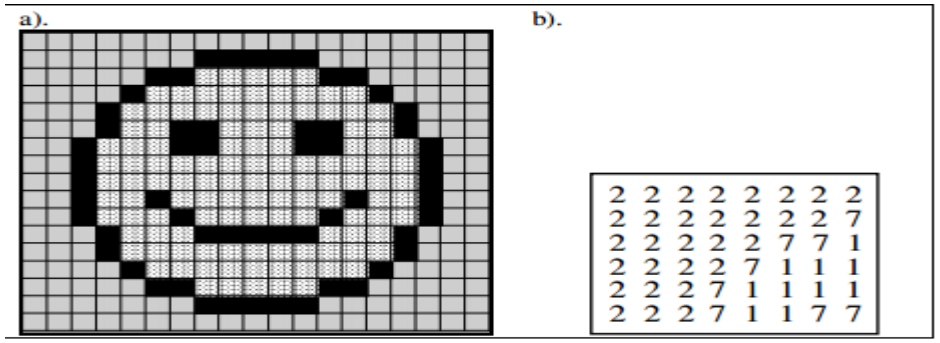
Raster Images

- Image is made up of many cells, in different shades.
- Each cell is known as pixel



Elements of Pictures Created in Computer Graphics

- Raster image:
- Stored in a computer as a 2D array of numerical values with a certain number of rows and a certain number of columns.
- Each numerical value represents the value of the pixel stored there.
- The array as a whole is often called a "**pixel map**". Or "**bitmap**"



Elements of Pictures Created in Computer Graphics

How are raster images created?

- **Hand designed images.**
  - A designer figures out what values are needed for each cell, and store it in memory
  - Paint program
- **Computed Images.**
  - An algorithm is used to “render” a scene, which might be modeled abstractly in computer memory.
- **Scanned images.**
  - A photograph or television image can be digitized

Elements of Pictures Created in Computer Graphics

Representation of gray shades and color in Raster Images:

- An important aspect of a raster image is the manner in which the various colors or shades of gray are represented in the bitmap.

  - Monochrome Raster images
  - Gray-scale Raster Images.
  - Color Raster images

Elements of Pictures Created in Computer Graphics

- **Representation of gray shades and color in Raster Images:**
  - **Monochrome raster images**
    - Pixels can contain one of two values in a raster image(bi-level)



Elements of Pictures Created in Computer Graphics

Representation of gray shades and color in Raster Images:

- **Gray-scale Raster Images.**
  - Pixel contains a gray color from several colors depending on pixel depth.
  - Pixel depth(number of bits needed to represent grey level)
    - N bit has 2^n possible color values
      - 2 bits/pixel produce 4 gray levels
      - 4 bits/pixel produce 16 gray levels
      - 8 bits/pixel produce 256 gray levels



2 bit



4 bit



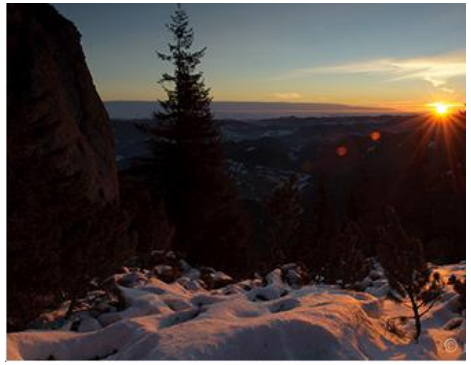
8 bit

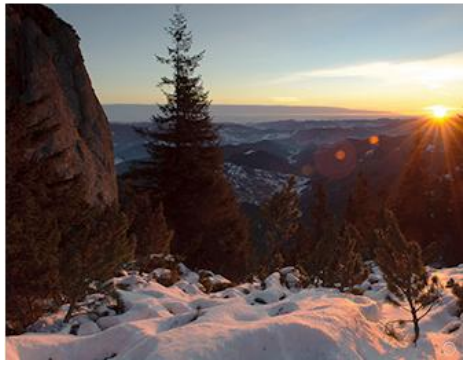
Elements of Pictures Created in Computer Graphics

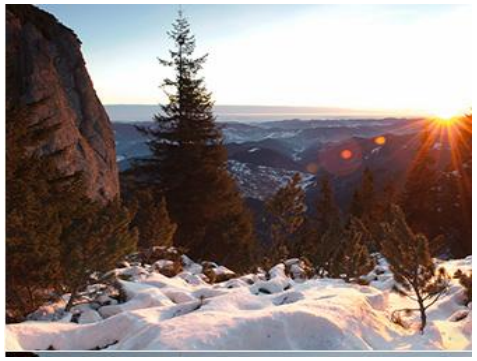
Representation of gray shades and color in Raster Images:

- **Color Raster Images.** There are several ways to associate a color, but one of the most common is a combination of red, green, and blue light.
  - Pixel contains a triple color of red, green and blue value
  - Color depth(number of bits needed to represent color of each pixel)
    - Color depth of 3 allows one bit for each color
      - True color image has a color depth of 24 bit(8 bit for each color)

color value	displayed
0,0,0	black
0,0,1	blue
0,1,0	green
0,1,1	cyan
1,0,0	red
1,0,1	magenta
1,1,0	yellow
1,1,1	white

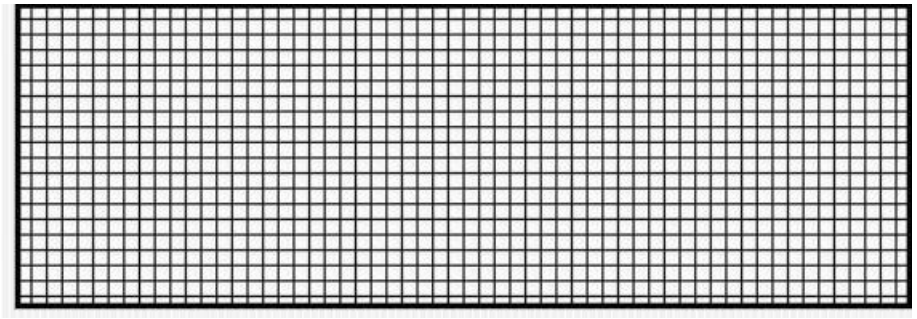






Raster Displays.

- Framebuffer is a portion of memory containing data representing all the pixels in a complete video frame(A rectangular array of points)
- Pixel : one dot of the raster
- Scan line: A row of pixels
- Example





Video Monitor



Flat panel



Laser printer

Graphics Display devices

- The display device is an output device used to represent the information in visual form.
  - Most displays used today for computer graphics are raster displays. The most familiar raster displays are **flat panel** which include:
    - Smartphones and Tablets
    - Smart TVs
    - Digital Signage
    - Automotive Infotainment Screens

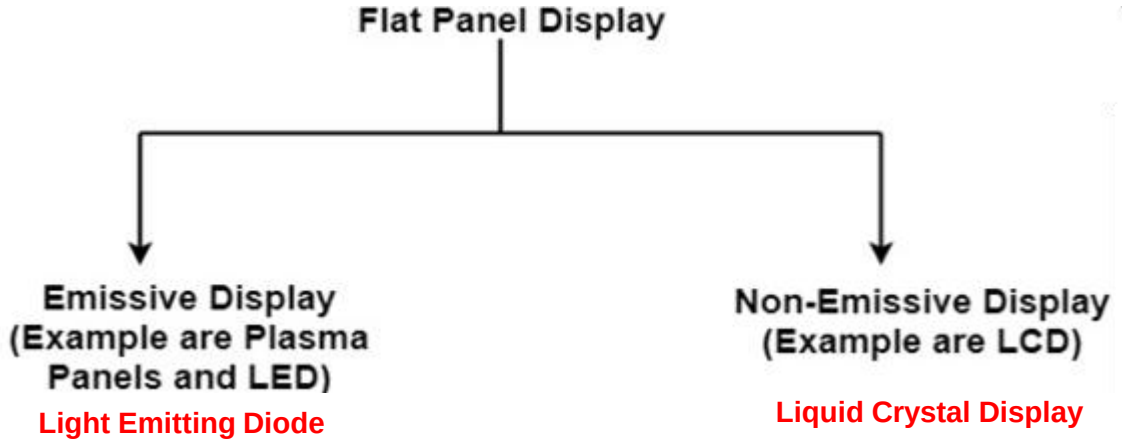
Types of Flat display: We can separate flat-panel displays into two categories

Emissive displays:

- The emissive displays (or emitters) are devices that convert electrical energy into light.

Non-emissive displays:

- use optical effects to convert light from some other source into graphics patterns.



Frame Rate vs. Refresh Rate & Matching

•**Frame Rate (FPS - Frames Per Second):** How many unique, complete images your **GPU (Graphics Processing Unit)** can create and send to the monitor each second. This is a **variable** number that depends on the power of your device.

•**Refresh Rate (Hz - Hertz):** How many times per second your **Monitor** can physically update the image it displays. This is a **fixed, maximum** number determined by the monitor's hardware (e.g., 60Hz, 144Hz).
"If one is higher or lesser than the other, the only thing that will happen is that the lesser one will put a cap on the higher one.“

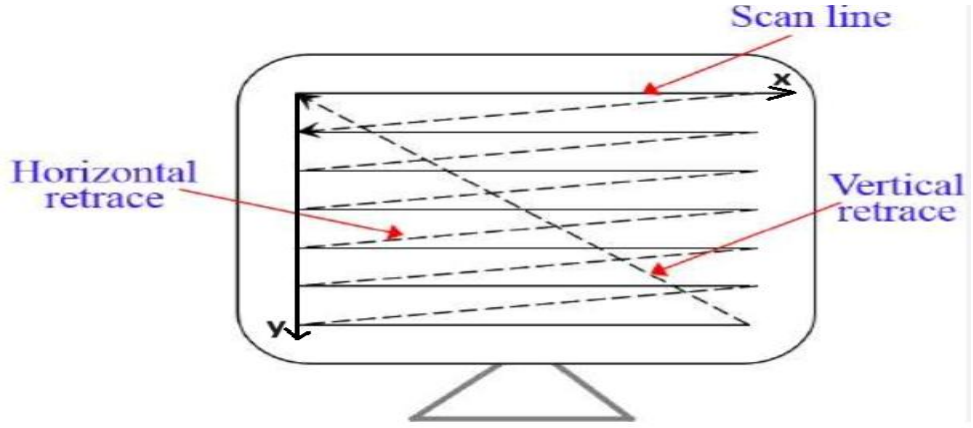
- **GPU Frame Rate < Monitor Refresh Rate (e.g., 45 FPS on a 60Hz monitor):**
  - The monitor is ready for a new frame, but the GPU hasn't finished creating one.
  - The monitor will simply **display the *old* frame again**, leading to **stuttering** or a **juddery** feel. The **lower frame rate caps the experience.**
- **GPU Frame Rate > Monitor Refresh Rate (e.g., 100 FPS on a 60Hz monitor):**
  - The GPU is creating frames faster than the monitor can show them.
  - The monitor can only display 60 of those 100 frames per second. This can cause **screen tearing**, where parts of two or more frames are shown on the screen at once because the monitor started drawing a new frame in the middle of its cycle.

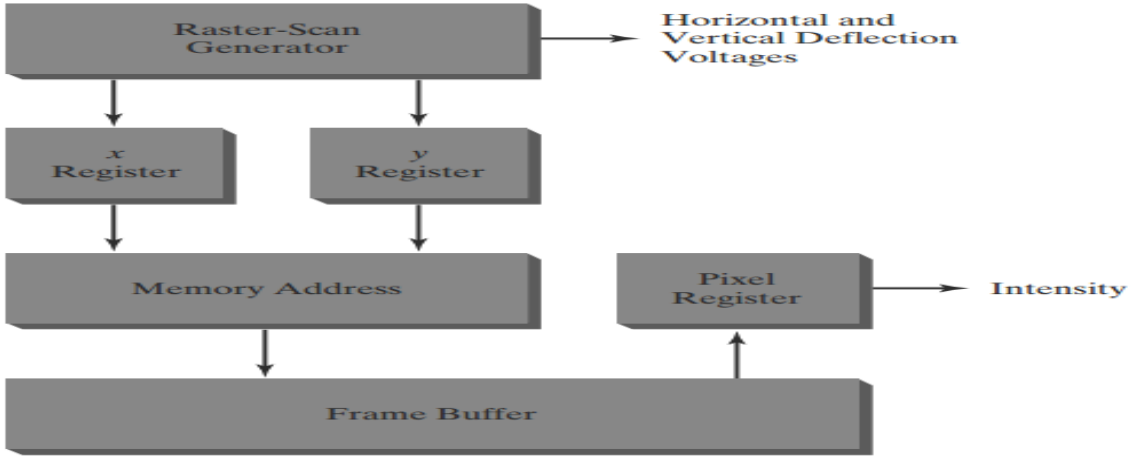


"Therefore, you should match the two features."

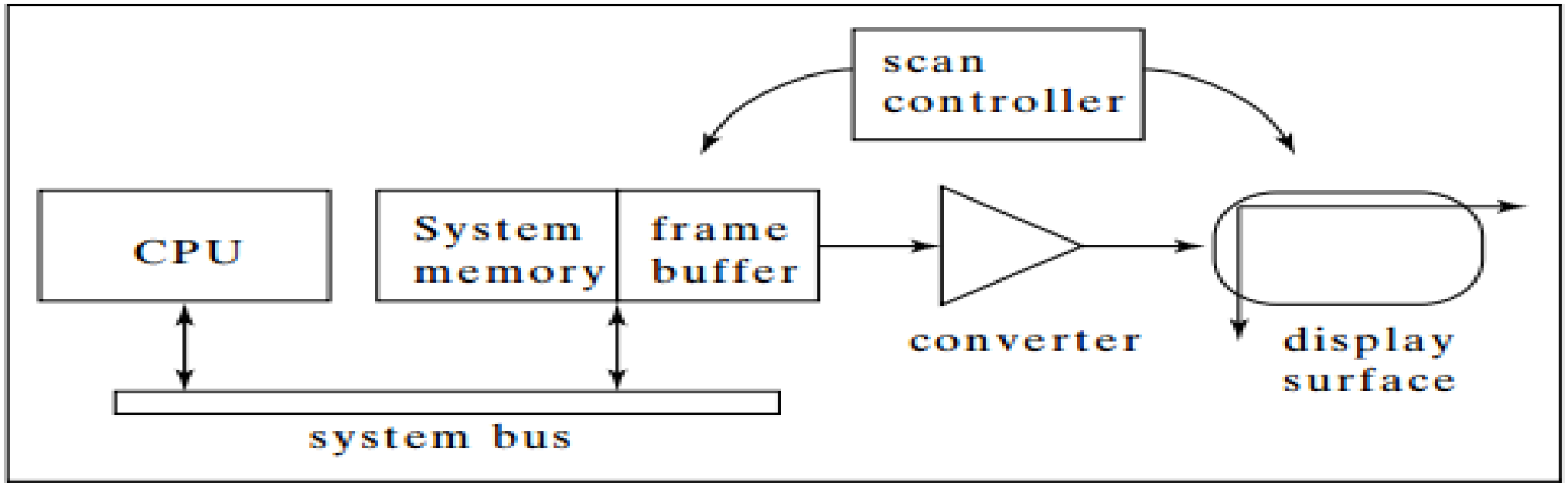
Raster Scan Displays

- Raster devices have a **display surface** on which the image is presented.
- The display surface has a certain number of pixels that it can show.
  - such as 480 rows, where each row contains 640 pixels.
    - In old raster scan system, the electron beam is swept across the screen one row at a time.
    - Nowadays Instead of an electron beam, the entire frame of pixel data is sent from the graphics card to the monitor digitally (via HDMI, DisplayPort, etc.).This data is sent as a continuous stream, row by row, in a process that is the direct digital descendant of the old raster scan.



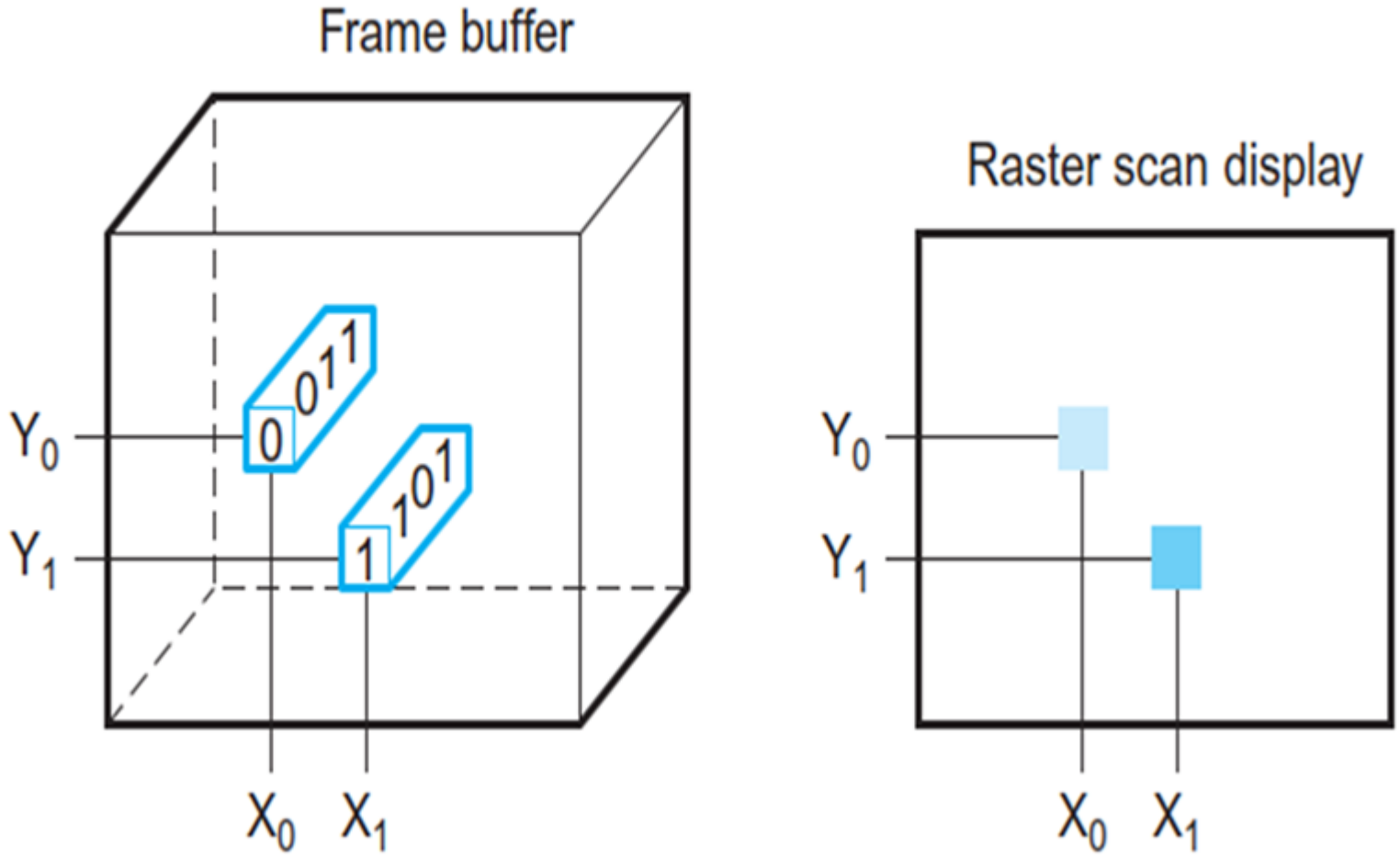


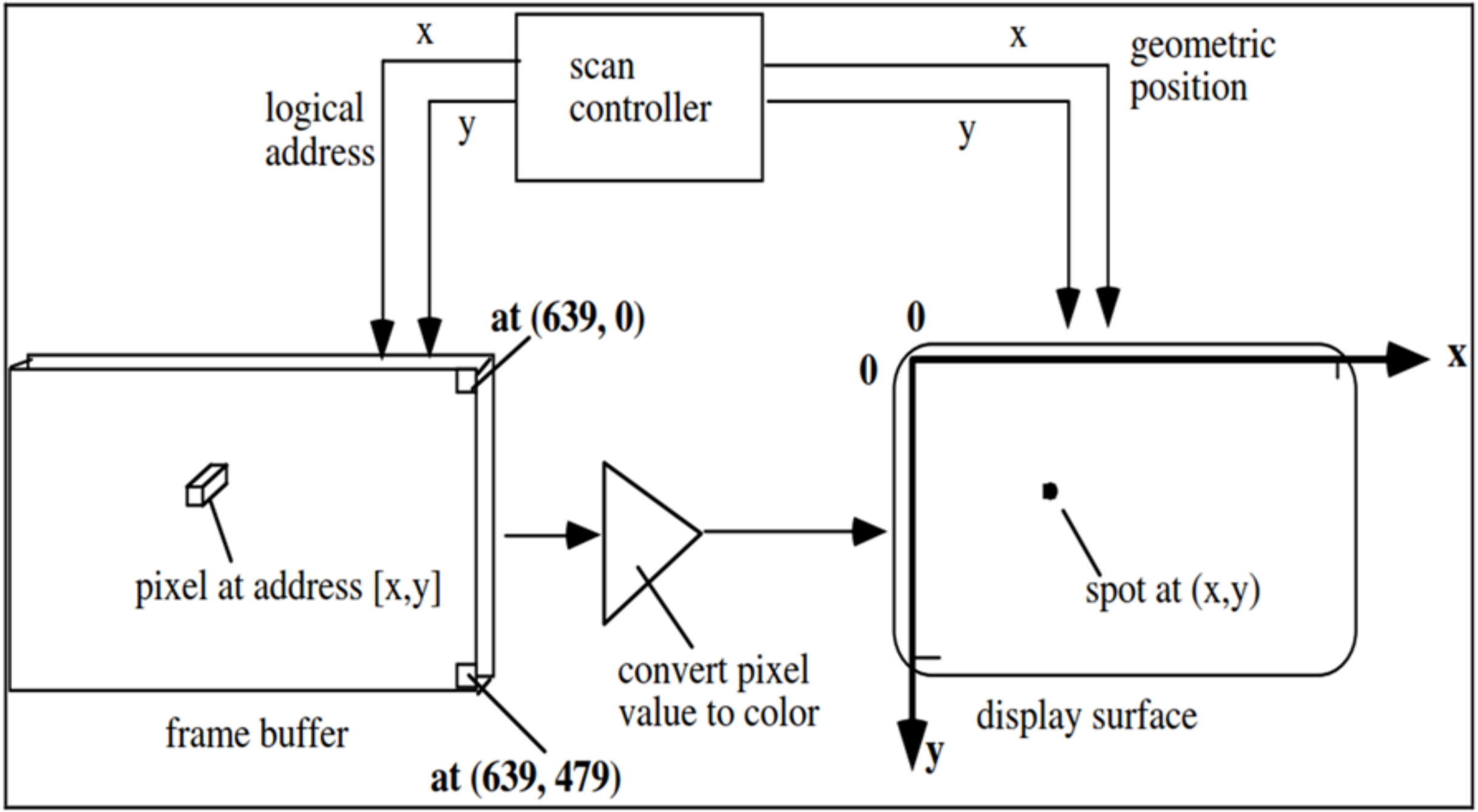
Raster scan system



- **The graphics program** is stored in system memory and is executed by (CPU).
- **Framebuffer** containing all the pixels intensity value in a complete video frame
- **scan controller** takes care of the actual display process. It causes the frame buffer to “send” each pixel through a converter to the appropriate physical spot on the display surface.
- **The converter** takes a pixel value such as 01001011 and converts it to the corresponding quantity that produces a spot of color on the display.

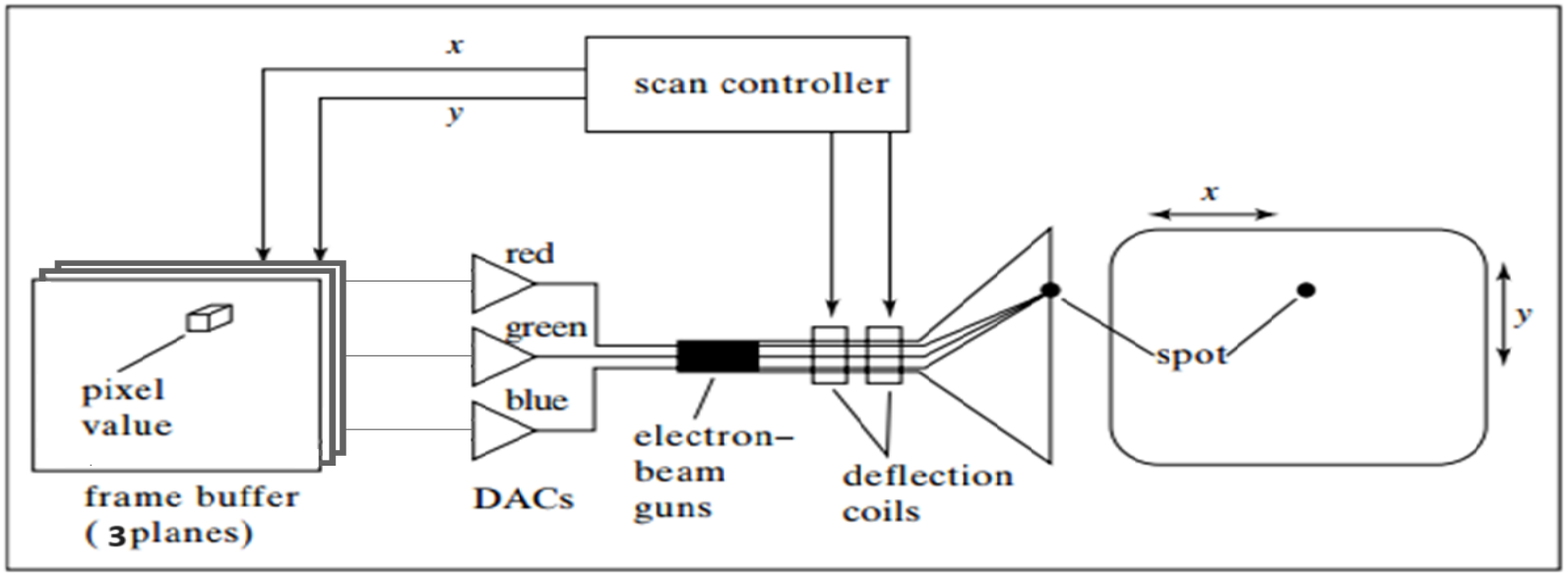
Raster scan system




x
logical
address
y
scan
controller
x
geometric
position
y
at (639, 0)
0
0
x
pixel at address [x,y]
convert pixel
value to color
spot at (x,y)
frame buffer
display surface
y
at (639, 479)

Video Monitors.

- Video monitors are based on a **CRT**, or cathode-ray tube, like the display in a television set
- For a simple color, frame buffer is implemented with 3 bit planes, one for each primary color(red, green, blue) Each bit plane drives an individual color gun.
- The deflection coils divert the three beams, so they stimulate three tiny phosphor dots at the proper place (x, y) on the inside of the cathode ray tube.
- One dot glows red when stimulated, one glows green, and one glows blue.
- The dots are so close together your eye sees one composite dot of a perceives color that is the sum of the three component colors.

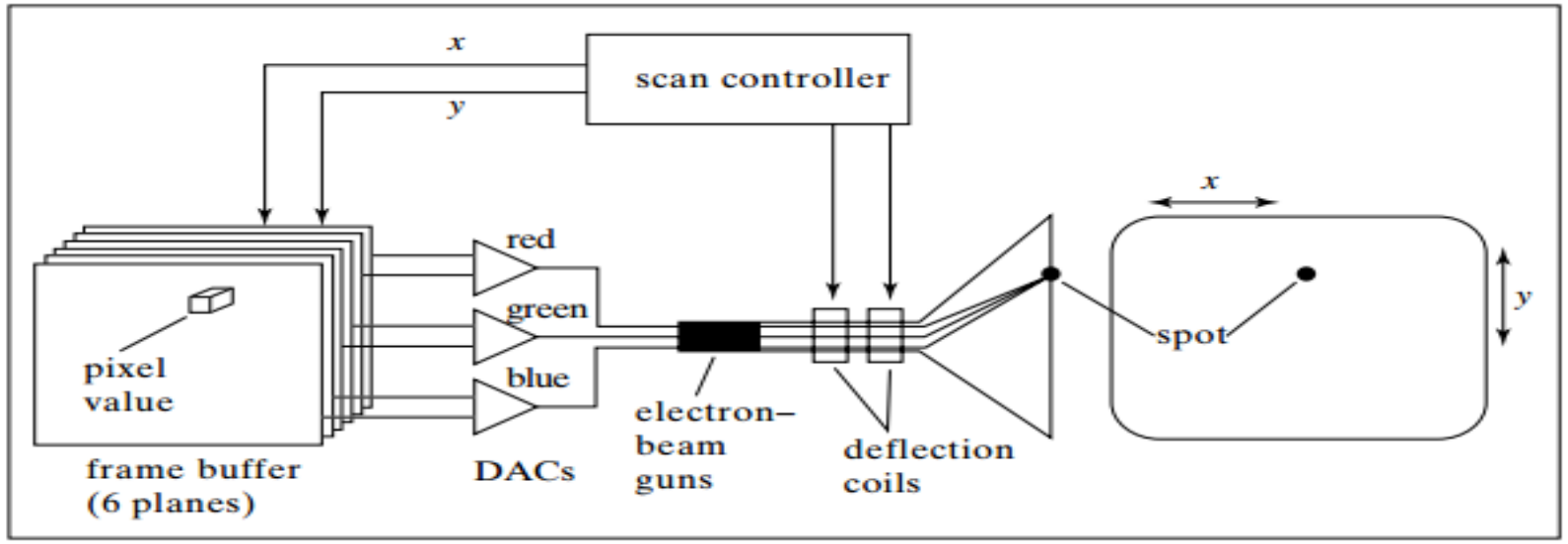


Red	Green	Blue	Color
0	0	0	Black
0	0	1	Blue
0	1	0	Green
0	1	1	Cyan
1	0	0	Red
1	0	1	Magenta
1	1	0	Yellow
1	1	1	White

Video Monitors.

For better resolution, additional bit planes can be used for each of the three-color. Thus, the composite dot can be made to glow in a total of $2^2 x\ 2^2 x\ 2^2 = 2^6 = 64$ different colors.

To normalize output voltage from 0 to max you need to divide each pixel intensity with its max



input	voltage/brightness
00	0 * Max
01	0.333 * Max
10	0.666 * Max
11	1 * Max

Input-output characteristic of a two-bit DAC.

Pixel color representation (Color palette)

- Color palette is the set of possible colors displayed by the system.

1. Direct coding → represents each pixel's color directly

For example:

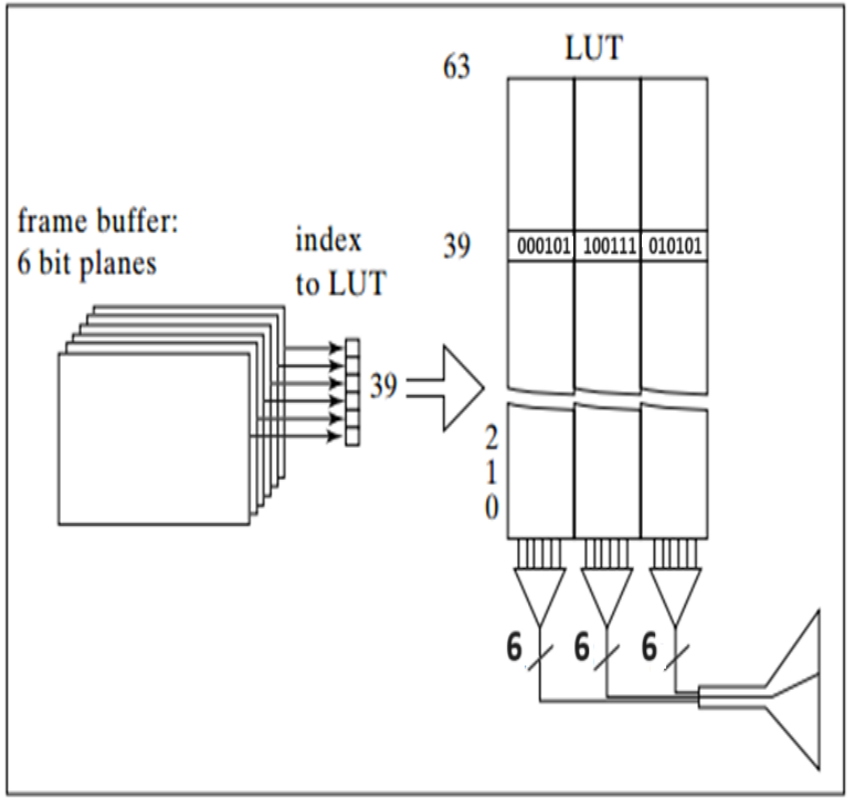
- Frame buffer that supports 6 planes of memory(2 planes for each color)
- Each of the DAC's has 2 input bits 2^2=4.
- So, there are 4 levels of red, 4 of green, and 4 of blue→ for a total of 2^6 color = 64 colors.
- It has a palette of 64 colors

- Frame buffer that supports True color image (a color depth of 24 bit)
- Which mean 24 planes of memory(8 planes for each color)
- Each of the DAC's has eight input bits 2^8=256.
- So, there are 256 levels of red, 256 of green, and 256 of blue→ for a total of 2^{24} color = 16 million color
- It has a palette of 16 million colors.

Pixel color representation (Color palette)

2. Indexed Color and LookUp Table (LUT)→ offers a **programmable association** between pixel value and final color which make the small bit per pixel to be almost good as high bit per pixel

- LUT is a compromise between our desire to have a lower storage requirement and our need to support enough number of colors.
- The color information is not directly carried by the image pixel data, but is stored in a separate piece of data called a color lookup table (CLUT) or palette.
- Every element in the array represents a color, indexed by its position within the array.
- Each image pixel does not contain the full specification of its color, but only its index into the palette.



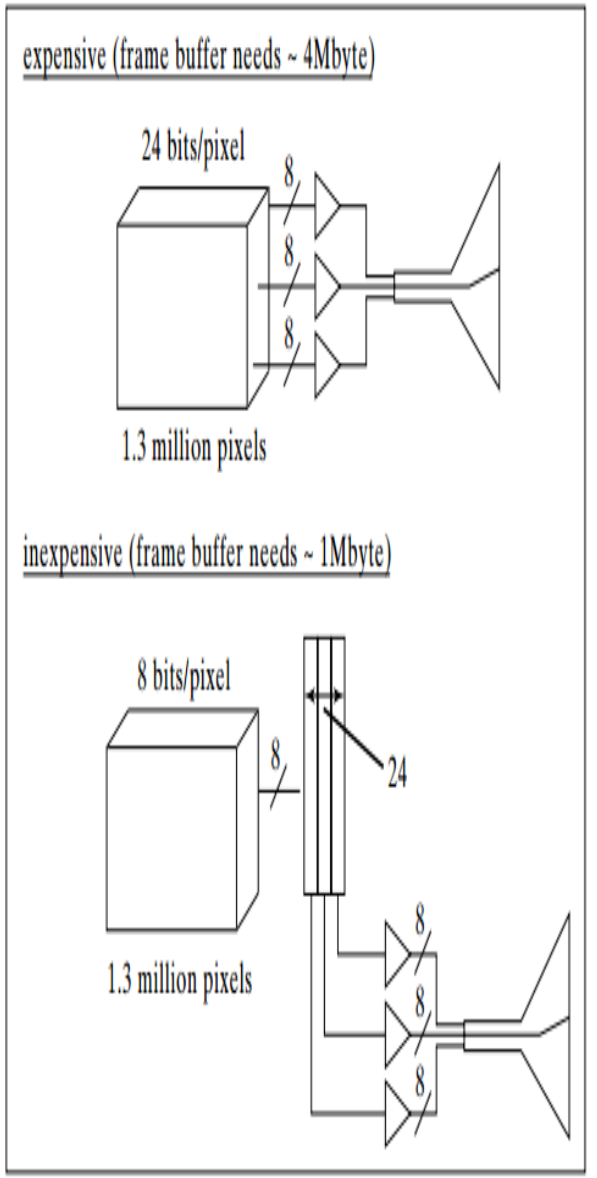
Indexed Color and the LUT.

- setPalette(39, 17, 25, 4)
- To make a particular pixel glow in this color, say the pixel at location (x, y) = (479, 532),
- the value 39 is stored in the frame buffer, using drawDot() defined earlier: drawDot(479, 532, 39);

System #1 (expensive): The first has a 24 bit/pixel frame buffer and no LUT, so each of its 1.3 million pixels can be set to any one of 2^{24} colors. Eight bits drive each of the DAC's. (The number '8' and the slash through the line into each DAC indicate the presence of eight bit lines feeding into the DAC.) The amount of memory required for the frame buffer here is 1024 x 1280 x 24 bits which is almost 4 megabytes.

System #2 (inexpensive): The second has an 8 bit/pixel frame buffer along with a LUT, and the LUT is 24 bits wide. The system can display 2^{24} different colors, but only 256 at a time. The amount of memory required for the frame buffer here is 1024 x 1280 x 8 which is about 1 megabyte. (The LUT requires a trivial 768 bytes of memory.) If memory costs a significant amount per megabyte, this system is much less expensive than system #1.

LUT OF (2^8*24)/8=768 BYTES



- If we use direct coding of RGB value with 3 bits per primary color, how many possible colors do we have for each pixel?
  - Solution →
    - $2^3 \times 2^3 \times 2^3 = 512$ color

- If we use 10 bits pixel value in a 24-bit lookup table representation, how many entries does the lookup tables have and how much is the size of the lookup table in bytes?
  - Solution →
    - lookup tables entries= $2^{10} = 1024$ entry
    - lookup table size = $(2^{10} \times 24)/8 = 3072$ byte

Thank You